Switching Circuits Laboratory (CS29002)

Assignment 1

January 7-14, 2026

Assignment 1 (to be completed in 2 lab days)

There are several parts in this assignment.

a) Realize a NOT gate (inverter) using the IC chip 4007. This chip provides three nMOS and three pMOS transistors.

b) Study the inverter's transfer characteristics by varying the input voltage in steps and measuring both the input and output voltages with a DSO. Plot the readings, and obtain the high and low noise margins for the gate.

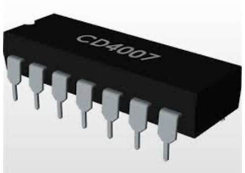
c) Realize a 3-input CMOS NAND gate and verify the truth table.
Realize a 3-input CMOS NOR gate and verify the truth table.

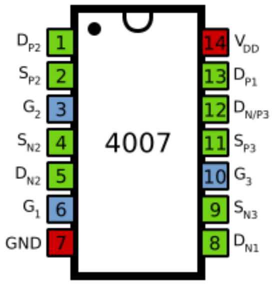
d) Design a ring oscillator by cascading an odd number N of NOT gates in series with feedback. Estimate the propagation delay of one NOT gate by measuring the frequency of the generated waveform on the DSO. Experiment with both TTL and CMOS NOT gates, for N=3 and N=5.

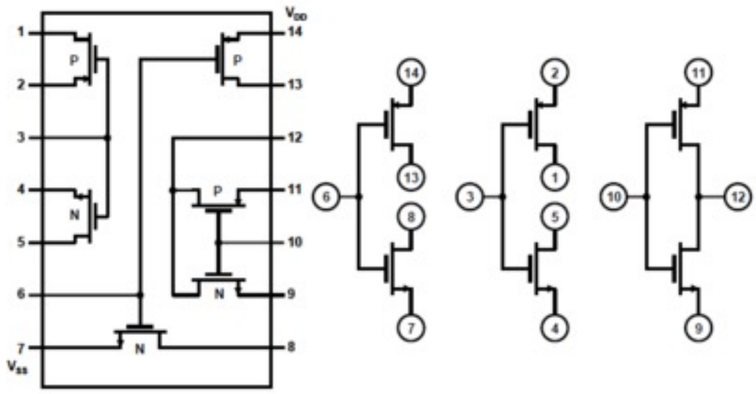
e) Implement a 4-bit (i) binary to Gray code converter, and (ii) Gray code to binary converter, using the TTL exclusive-OR IC 7486. Apply the inputs using on-board switches, observe the outputs on on-board LEDs, and verify the truth table.

The IC Chip 4007

- Contains three nMOS transistors and three pMOS transistors.

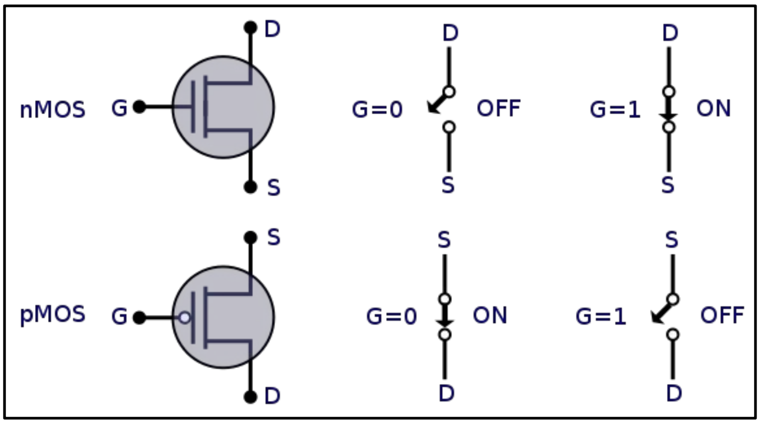






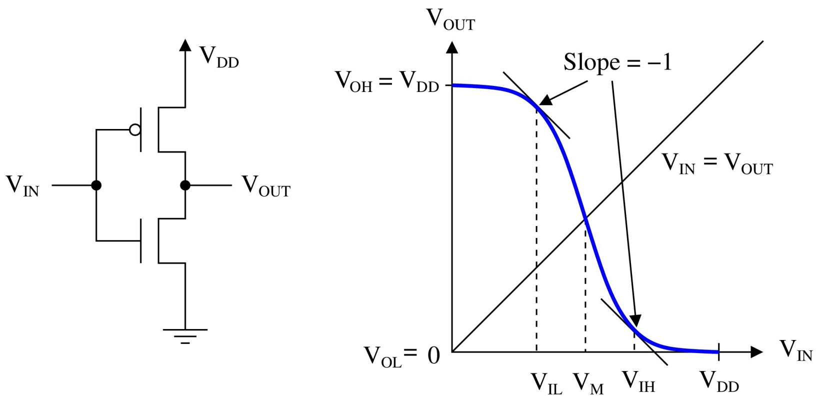
3

Symbolic Notations of nMOS and pMOS transistors



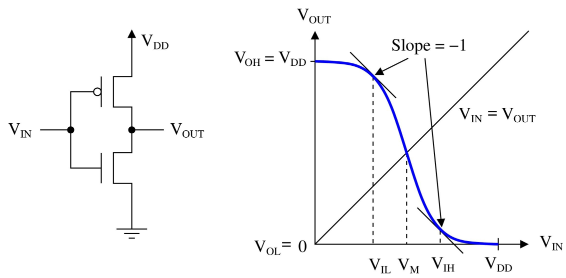
4

CMOS Inverter (NOT Gate)



(a) (b)

5



(a) (b)

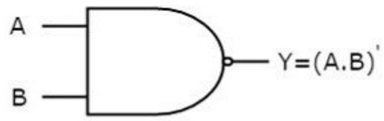
Part (b) shows the typical DC characteristics of a CMOS inverter. The input-high and input-low voltages are usually taken to be those input voltages at which the slope of the curve is -1. The output-high and output-low voltages are V_{DD} and 0. The high and low noise margins are defined as:

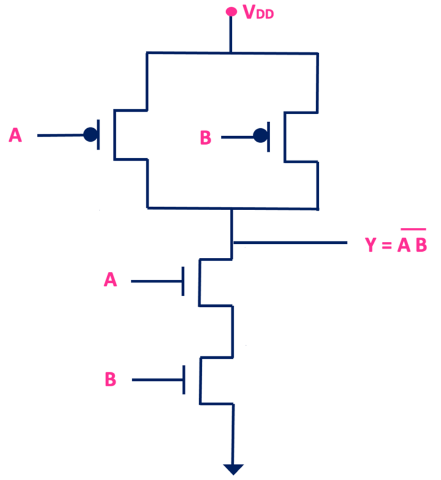
$NM_H = V_{OH} – V_{IH} = V_{DD} – V_{IH}$

$NM_L = V_{IL} – V_{OL} = V_{IL}$

6

2-input CMOS NAND Gate



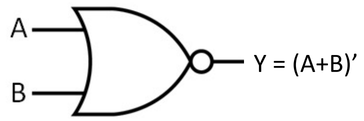


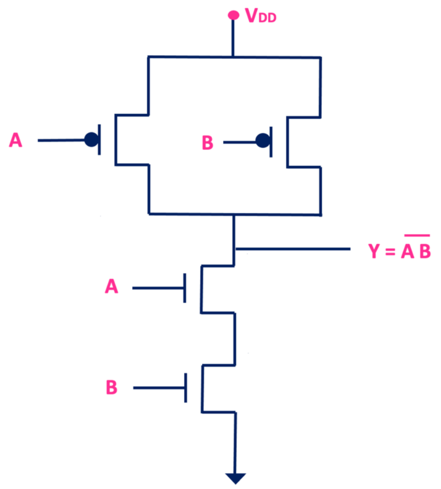
Truth Table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

7

2-input CMOS NOR Gate



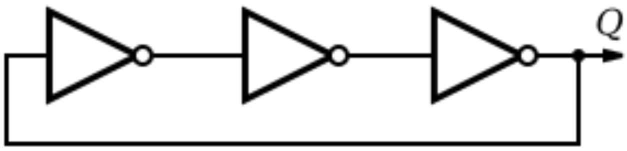


Truth Table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

8

Ring Oscillator using 3 NOT Gates

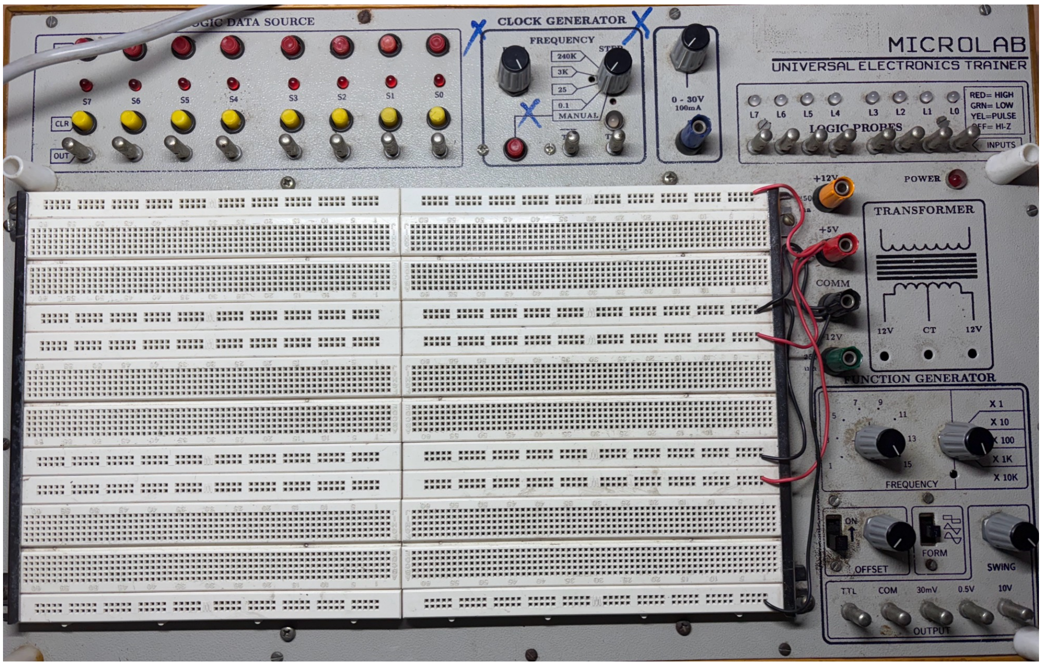


If *T* is the time delay of a single inverter, and *N* is the number of inverters used in the ring oscillator (here, $N = 3$), the frequency of oscillation will be $f = 1 / (2NT)$.

Important Instructions

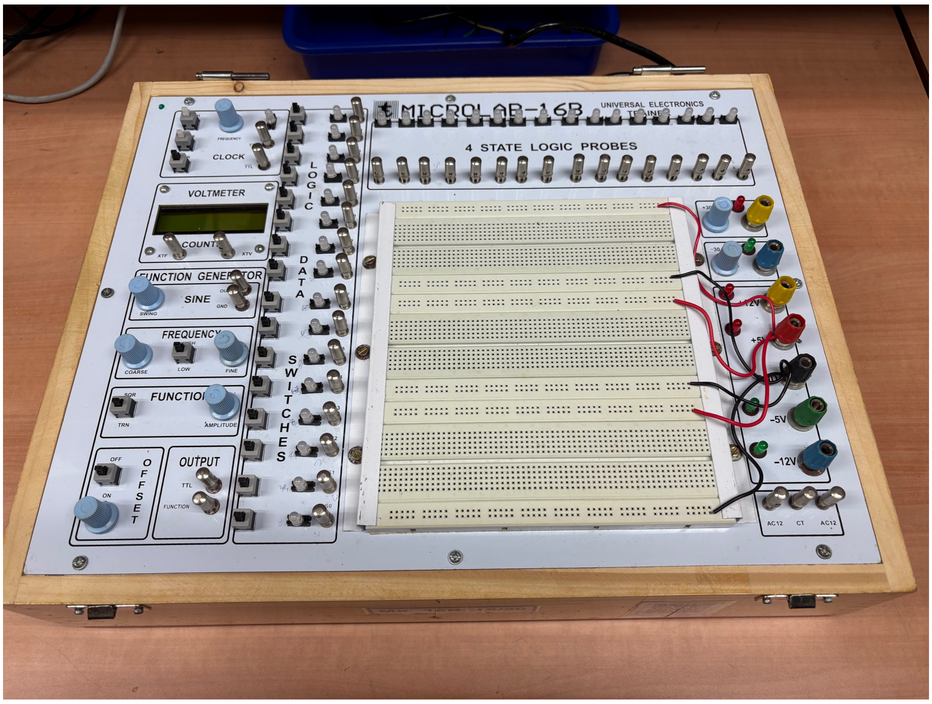
- For each design, draw the circuit diagram on paper, following the suggested conventions, and get it approved by one of the TAs. Then you show the diagram and issue the components.
- After you finish each part of the assignment, show it to your assigned TA, who will be doing the evaluation.
- Every group is expected to bring a wire cutter and a twizzer to the lab. Special credit will be given for neatly completed circuit realizations.
- You must prepare a laboratory report for each experiment *in PDF format* and upload it to Moodle within the specified deadlines. A single report has to be uploaded per group.

TYPE 1 Prototyping Kit



11

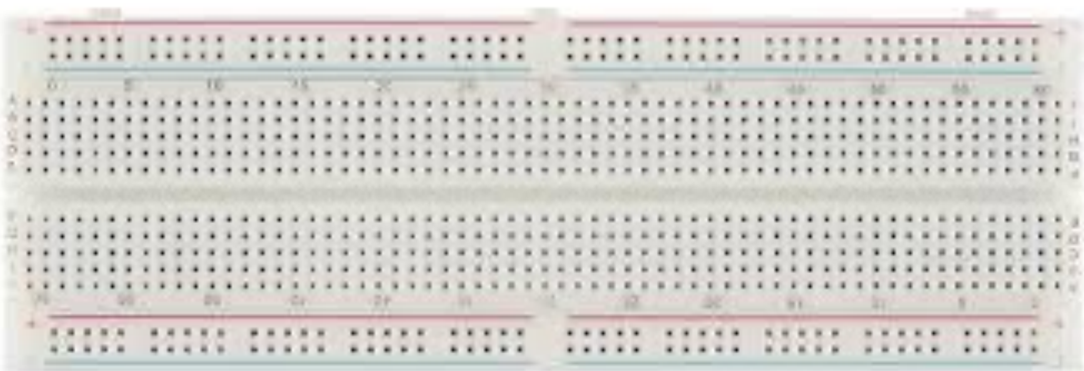
TYPE 2 Prototyping Kit



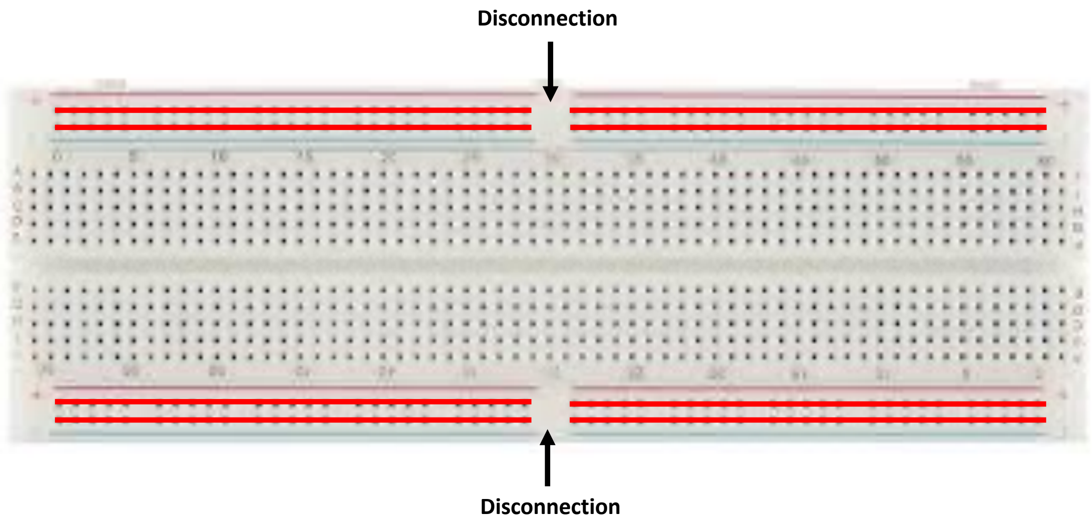
12

Wire Cutter and Twizzer

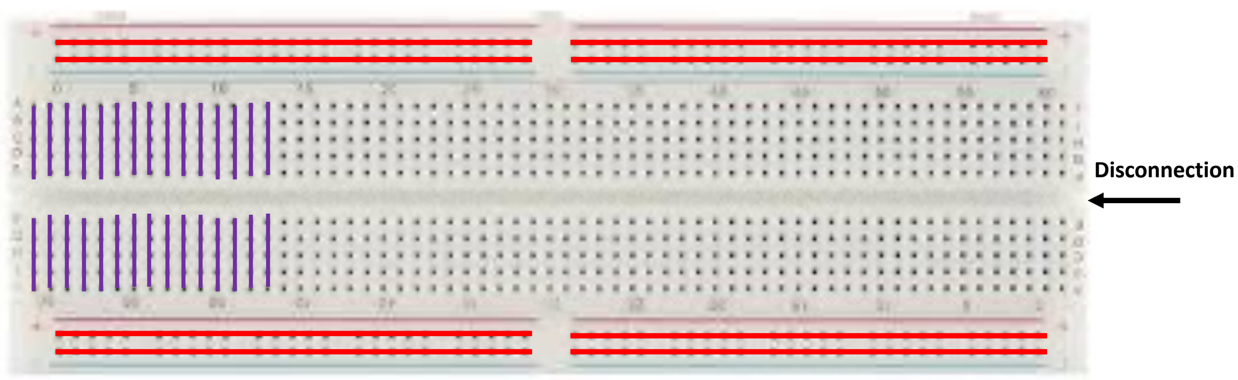
13



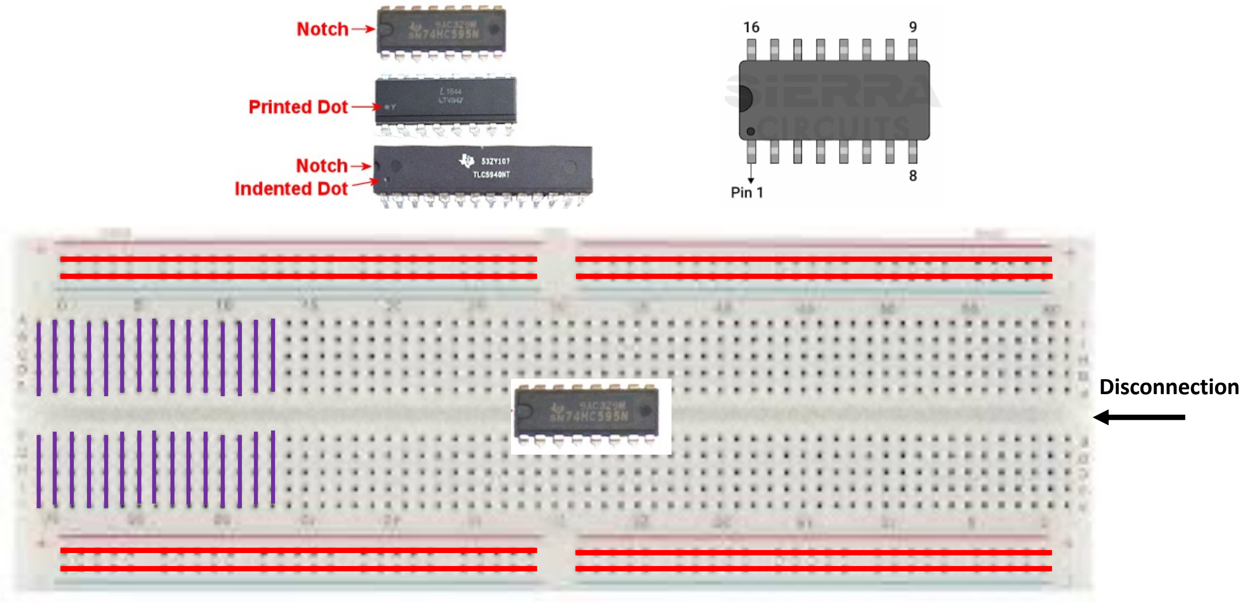
14



15

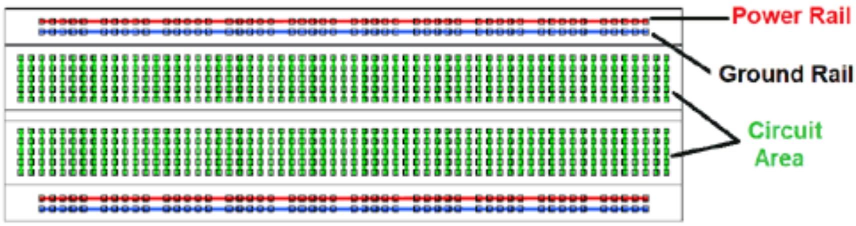


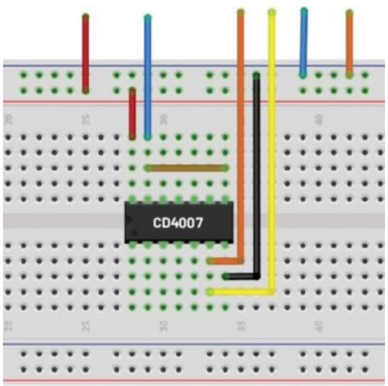
16



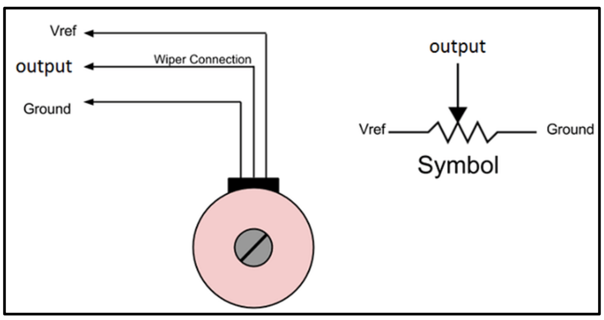
17

Default connections on breadboard





Typical connections on breadboard



Variable voltage source using potentiometer

18

Thank You

19